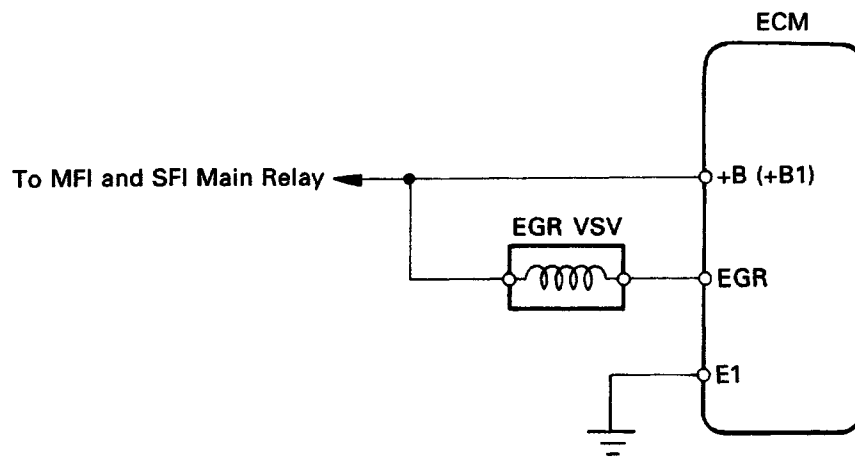


EGR VSV

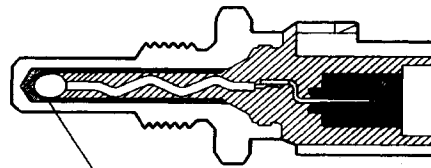


P01447

INSPECTION OF EGR VSV

4A-FE (See page [EC-11](#))3S-GTE (See page [EC-23](#))5S-FE (See page [EC-38](#))

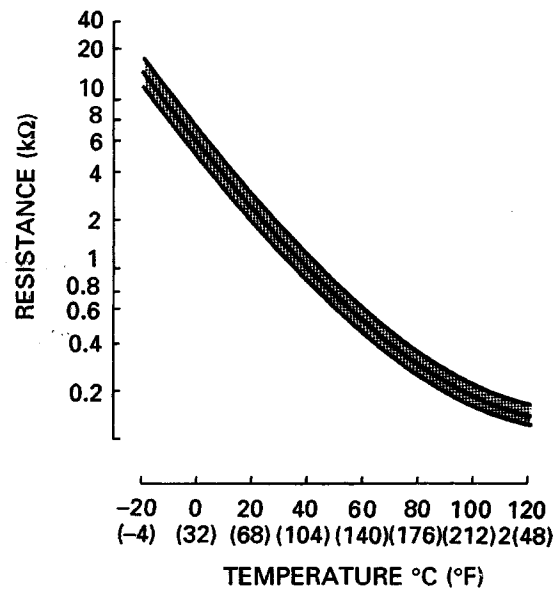
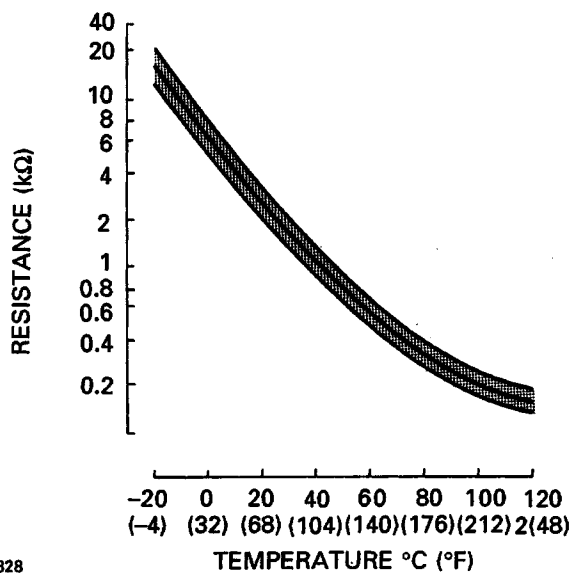
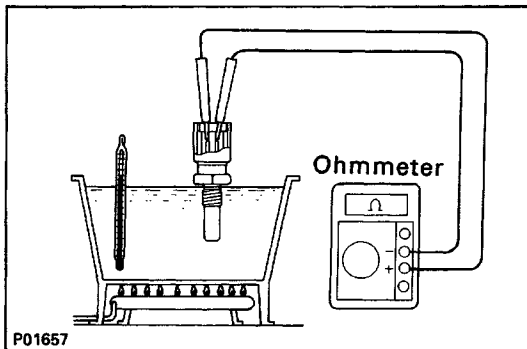
Engine Coolant Temperature Sensor



Thermistor

4A-FE

3S-GTE and 5S-FE

FI3828
P03525 P03524

P01657

INSPECTION OF ENGINE COOLANT TEMPERATURE SENSOR

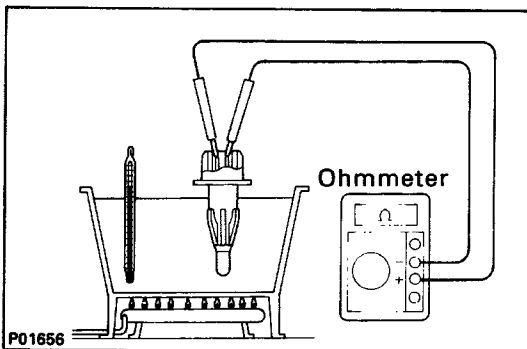
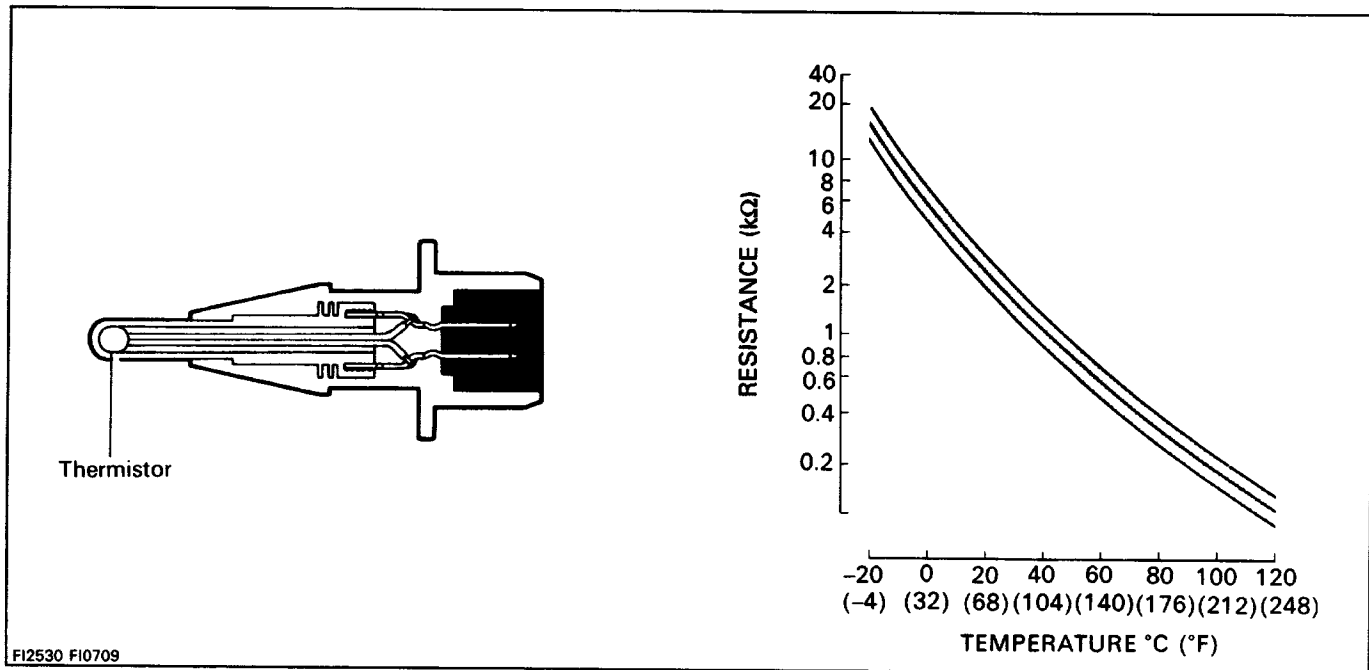
INSPECT RESISTANCE OF ENGINE COOLANT TEMPERATURE SENSOR

Using an ohmmeter, measure the resistance between the terminals.

Resistance: Refer to the chart above

If the resistance is not as specified, replace the sensor.

Intake Air Temperature Sensor (4A-FE and 5S-FE)



INSPECTION OF INTAKE AIR TEMPERATURE SENSOR

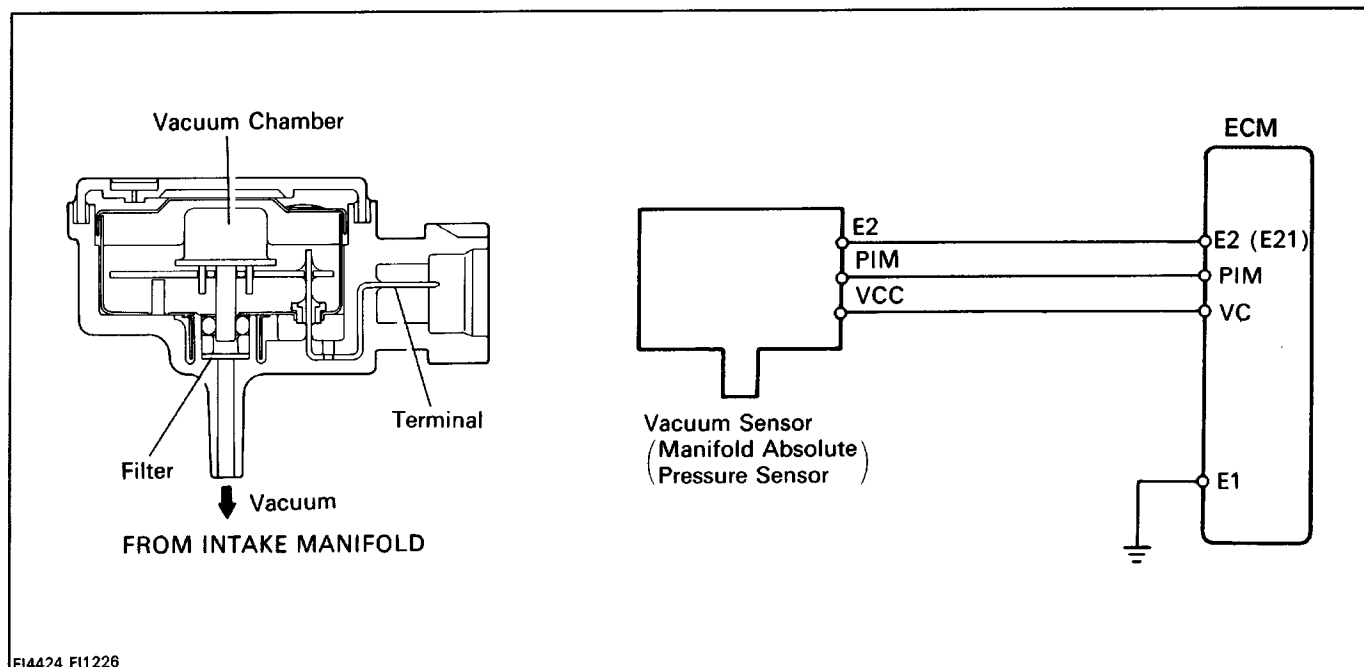
INSPECT RESISTANCE OF INTAKE AIR TEMPERATURE SENSOR

Using an ohmmeter, measure the resistance between the terminals.

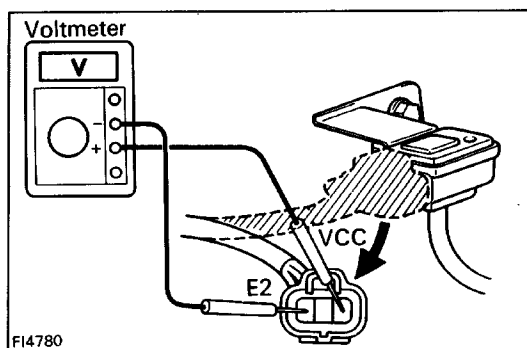
Resistance: Refer to the chart above

If the resistance is not as specified, replace the sensor.

Vacuum Sensor (Manifold Absolute Pressure Sensor) (4A-FE)



FI4424 FI1226



FI4780

INSPECTION OF VACUUM SENSOR

1. INSPECT POWER SOURCE VOLTAGE OF VACUUM SENSOR

- Disconnect the vacuum sensor connector.
- Turn the ignition switch ON.
- Using a voltmeter, measure the voltage between connector terminals VCC and E2 of the wiring harness side.

Voltage: 4.5 – 5.5 V

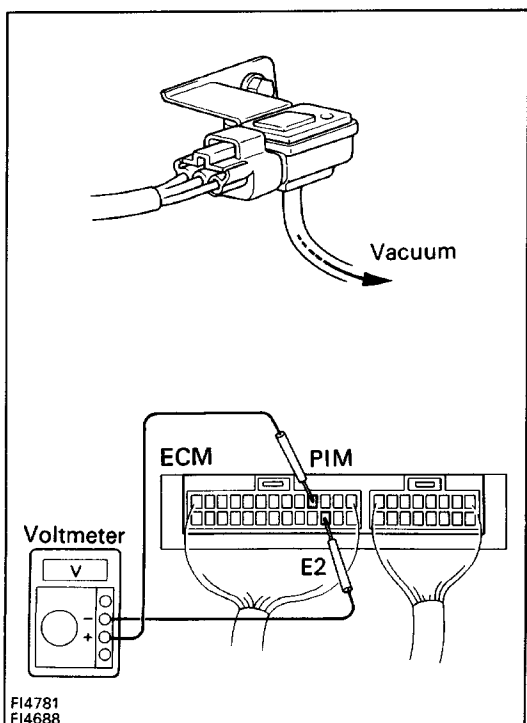
- Reconnect the vacuum sensor connector.

2. INSPECT POWER OUTPUT OF VACUUM SENSOR

- Turn the ignition switch ON.
- Disconnect the vacuum hose of the intake manifold (chamber) side.
- Connect a voltmeter to terminals PIM and E2 of the ECM, and measure and record the output voltage under ambient atmospheric pressure.
- Apply vacuum to the vacuum sensor in 13.3 kPa (100 mmHg, 3.94 in.Hg) segments to 66.7 kPa (500 mmHg, 19.69 in.Hg).
- Measure voltage drop from step (c) above for each segment.

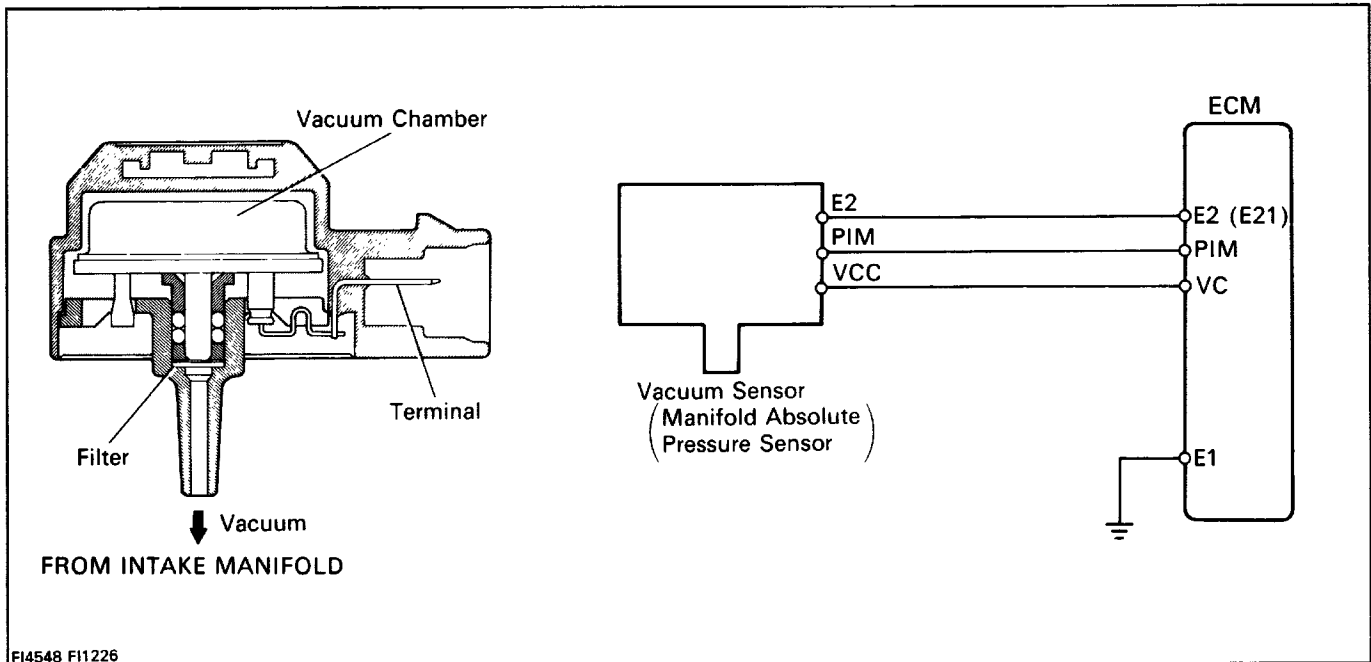
Voltage drop:

Applied Vacuum kPa (mmHg in.Hg)	13.3 (100 3.94)	26.7 (200 7.87)	40.0 (300 11.81)	53.5 (400 15.75)	66.7 (500 19.69)
Voltage drop (V)	0.3–0.5	0.7–0.9	1.1–1.3	1.5–1.7	1.9 – 2.1

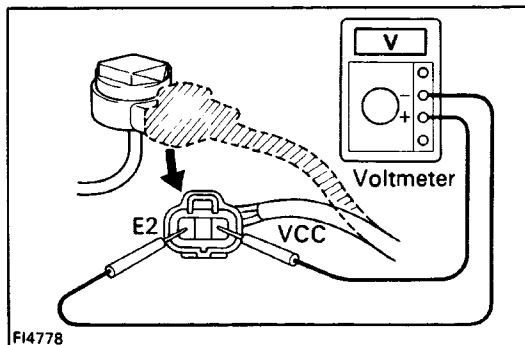


FI4781
FI4688

Vacuum Sensor (Manifold Absolute Pressure Sensor) (5S-FE)



FI4548 FI1226



FI4778

INSPECTION OF VACUUM SENSOR

1. INSPECT POWER SOURCE VOLTAGE OF VACUUM SENSOR

- Disconnect the vacuum sensor connector.
- Turn the ignition switch ON.
- Using a voltmeter, measure the voltage between connector terminals VCC and E2 of the wiring harness side.

Voltage: 4.5 – 5.5 V

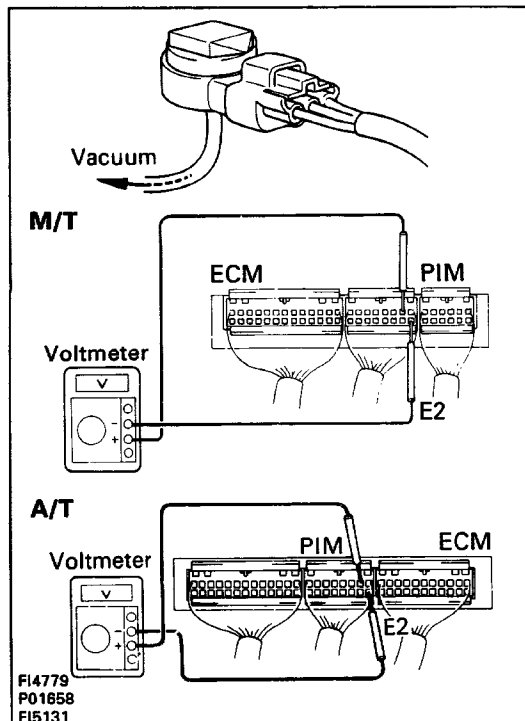
- Reconnect the vacuum sensor connector.

2. INSPECT POWER OUTPUT OF VACUUM SENSOR

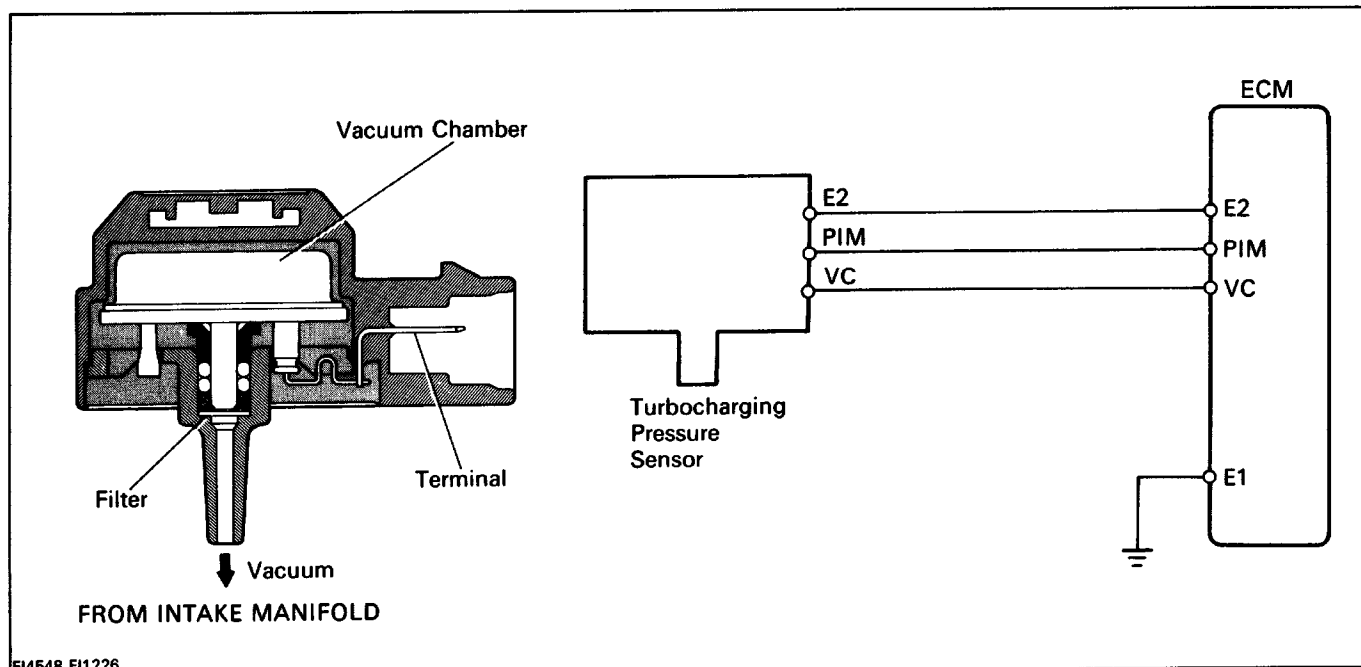
- Turn the ignition switch ON.
- Disconnect the vacuum hose of the intake manifold (chamber) side.
- Connect a voltmeter to terminals PIM and E2 of the ECM, and measure and record the output voltage under ambient atmospheric pressure.
- Apply vacuum to the vacuum sensor in 13.3 kPa (100 mmHg, 3.94 in.Hg) segments to 66.7 kPa (500 mmHg, 19.69 in.Hg).
- Measure voltage drop from step (c) above for each segment.

Voltage drop:

Applied Vacuum kPa mmHg in.Hg	13.3 (100 3.94)	26.7 (200 7.87)	40.0 (300 11.81)	53.5 (400 15.75)	66.7 (500 19.69)
Voltage drop (V)	0.3–0.5	0.7–0.9	1.1–1.3	1.5 – 1.7	1.9 – 2.1

FI4779
P01658
FI5131

Turbocharging Pressure Sensor (3S-GTE)



FI4548 FI1226

INSPECTION OF TURBOCHARGING PRESSURE SENSOR

1. INSPECT POWER SOURCE VOLTAGE OF TURBOCHARGING PRESSURE SENSOR

- Disconnect the pressure sensor connector.
- Turn the ignition switch ON.
- Using a voltmeter, measure the voltage between connector terminals VC and E2 of the wiring harness side.

Voltage: 4.5 – 5.5 V

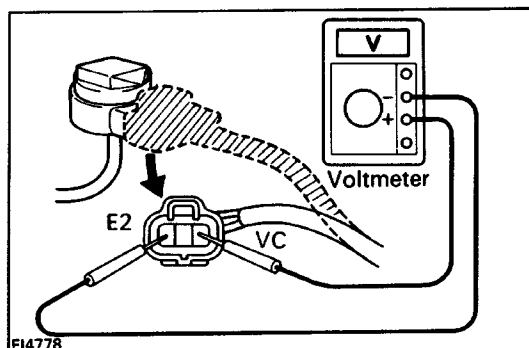
- Reconnect the pressure sensor connector.

2. INSPECT POWER OUTPUT OF TURBOCHARGING PRESSURE SENSOR

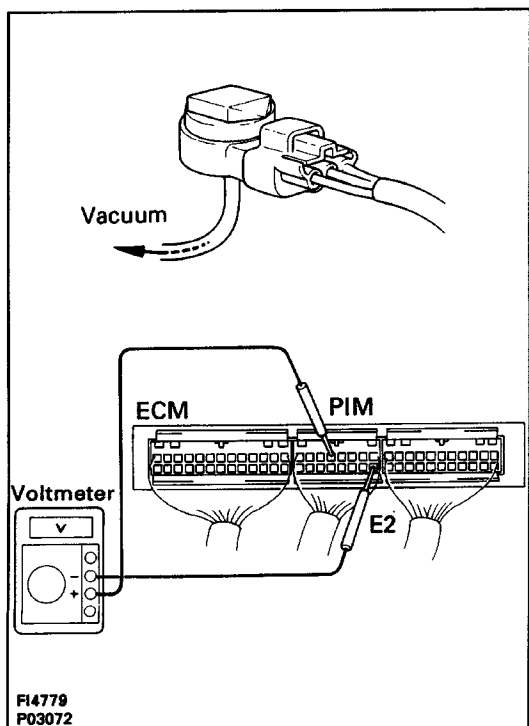
- Turn the ignition switch ON.
- Disconnect the vacuum hose of the intake manifold (chamber) side.
- Connect a voltmeter to terminals PIM and E2 of the ECM, and measure and record the output voltage under ambient atmospheric pressure.
- Apply vacuum to the pressure sensor in 13.3 kPa (100 mmHg, 3.94 in.Hg) segments to 66.7 kPa (500 mmHg, 19.69 in.Hg).
- Measure voltage drop from step (c) above for each segment.

Voltage drop:

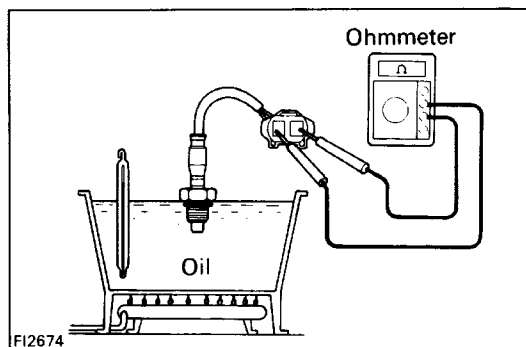
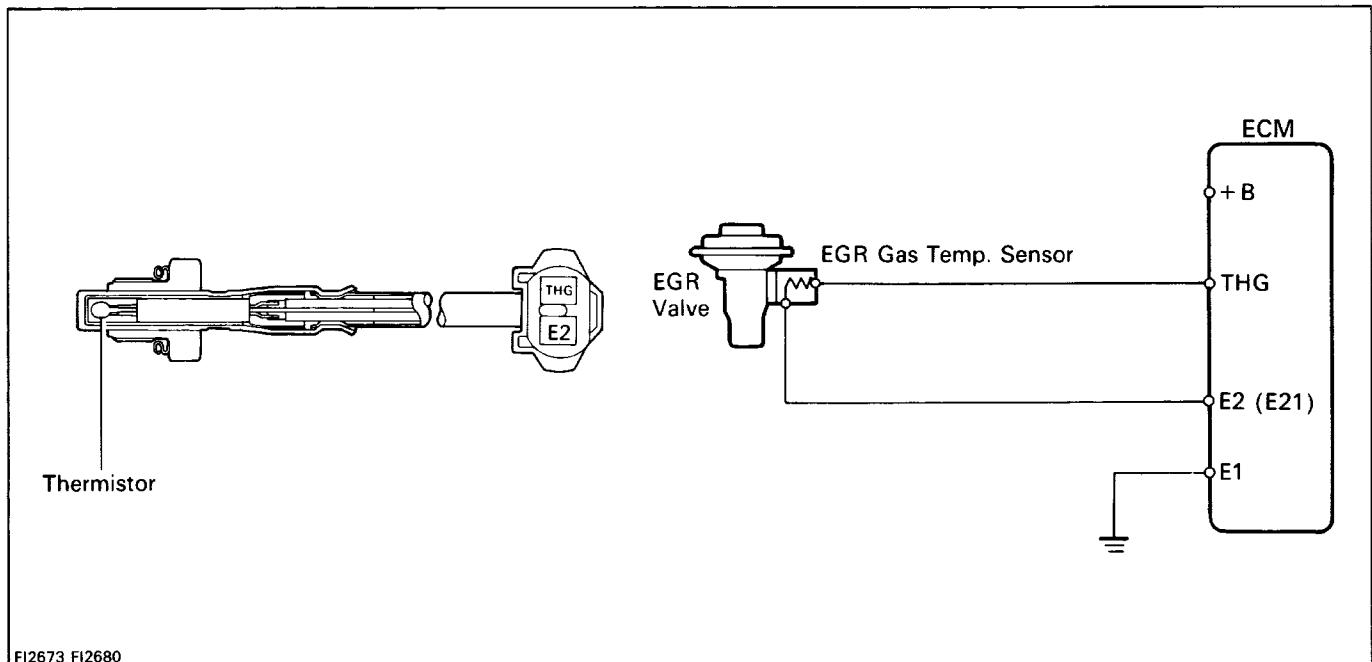
Applied Vacuum kPa	13.3	26.7	40.0	53.5	66.7
(mmHg in.Hg)	(100 3.94)	(200 7.87)	(300 11.81)	(400 15.75)	(500 19.69)
Voltage drop (V)	0.15–0.35	0.4–0.6	0.65–0.85	0.9–1.1	1.15–1.35



FI4778

FI4779
P03072

EGR Gas Temperature Sensor (CALIF. only)



INSPECTION OF EGR GAS TEMPERATURE SENSOR

INSPECT EGR GAS TEMPERATURE SENSOR

Using an ohmmeter, measure the resistance between the terminals.

Resistance:

69 – 89 k at 50°C (112°F)

11 – 15 k at 100°C (212°F)

2 – 4 k at 150°C (302°F)

If the resistance is not as specified, replace the sensor.